

INT374 – DATA ANALYTICS WITH POWER BI
PROJECT REPORT
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Major Irrigation Analysis

Submitted by

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CERTIFICATE

This is to certify that Kunal Kumar bearing Registration no.12310980 has completed INT374 project titled, “**Major Irrigation Analysis**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

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DECLARATION

I, Kunal Kumar student of Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

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1.INTRODUCTION

1.1 Background of the Study

In the present era of digital transformation, data has become one of the most valuable assets for organizations, governments, and researchers. With the rapid growth in data generation, there is a strong need to analyze, interpret, and visualize large datasets in an effective manner to support informed decision-making. Data analytics and visualization tools play a vital role in converting raw data into meaningful insights that can help stakeholders understand trends, patterns, and relationships within the data.

In sectors such as agriculture, irrigation management, and rural development, data-driven analysis is essential for planning, monitoring, and optimizing resources. Efficient irrigation management is critical for sustainable agricultural growth, especially in regions where water resources are limited. Analyzing irrigation-related data helps in understanding seasonal irrigation patterns, water usage, and the distribution of irrigation schemes across different geographical areas.

This project focuses on analyzing irrigation-related data using modern data analytics and visualization techniques. The study emphasizes transforming complex datasets into interactive dashboards that provide clear insights for better understanding and decision-making. The use of data visualization tools enhances the accessibility of information and supports stakeholders in evaluating performance and identifying improvement areas.

1.2 Problem Statement

Large volumes of irrigation and agricultural data are often available in raw and unstructured formats, making it difficult to extract meaningful insights directly. Traditional methods of data analysis lack interactivity and fail to provide a comprehensive view of seasonal trends, district-wise distribution, and irrigation coverage.

The absence of an integrated visualization platform makes it challenging for users to:

- Analyze irrigation patterns across different seasons
- Compare irrigated areas among districts and regions
- Identify key trends and anomalies in irrigation data

Therefore, there is a need for an interactive and visually intuitive dashboard that can process, analyze, and present irrigation data in a structured and user-friendly manner. This project aims to address these challenges by designing a comprehensive data model and an interactive dashboard for effective analysis.

1.3 Objectives of the Project

The primary objectives of this Minor Project are as follows:

- To understand and analyze irrigation-related datasets effectively
- To perform data preprocessing for cleaning and transforming raw data
- To design an efficient data model for analysis
- To create an interactive dashboard for visualizing irrigation data
- To analyze seasonal and district-wise irrigation patterns
- To extract meaningful insights that support data-driven decision-making

1.4 Scope of the Project

The scope of this project is limited to the analysis and visualization of the provided irrigation dataset. The project focuses on descriptive and exploratory data analysis using visualization techniques. It does not involve predictive modeling or real-time data integration.

The dashboard developed in this project enables users to:

- View irrigation data across different seasons
- Compare irrigated areas among districts and villages
- Interact with data using filters and slicers

The project is designed as a foundational analytical system that can be extended in the future for advanced analytics and automation.

1.5 Significance of the Study

This project highlights the importance of data analytics and visualization in understanding complex datasets related to irrigation management. By presenting data in an interactive dashboard format, the study simplifies data interpretation and enhances analytical efficiency.

The project demonstrates practical applications of data preprocessing, modeling, and visualization techniques, making it academically significant for students and researchers. Additionally, the insights derived from the dashboard can assist decision-makers in identifying patterns and areas that require attention, contributing to better planning and resource utilization.

2.DATASET DESCRIPTION

2.1 Source of the Dataset

The dataset used in this Minor Project has been collected from a structured and reliable source related to irrigation and agricultural data management. The data represents detailed information regarding irrigation schemes, irrigated areas, seasonal distribution, and groundwater levels across various administrative regions such as states, districts, blocks, and villages. The dataset serves as a comprehensive base for analyzing irrigation patterns and understanding the distribution of irrigation resources across different geographical units.

2.2 Overview of the Dataset

The dataset consists of multiple records, each representing irrigation-related information for a specific village or administrative unit. It includes both quantitative and qualitative attributes that capture geographical, agricultural, and irrigation-specific details. The dataset provides information about cultivable land, net sown area, gross irrigated area across different seasons, groundwater levels, and types of irrigation schemes.

The dataset is well-suited for analytical and visualization purposes as it enables comparison across regions and seasons. It also supports the identification of trends and patterns related to irrigation practices and water resource utilization.

2.3 Description of Attributes

The dataset contains several attributes that can be broadly categorized into geographical, agricultural, and irrigation-related variables. Some of the key attributes included in the dataset are:

- **State Name:** Represents the name of the state
- **District Name:** Indicates the district under the respective state
- **Block/Tehsil Name:** Specifies the administrative subdivision
- **Village Name:** Identifies individual villages
- **Geographical Area:** Total geographical area of the region
- **Cultivable Area:** Area suitable for cultivation
- **Net Sown Area:** Area actually sown with crops
- **Gross Irrigated Area (Season-wise):**
 - Kharif Season
 - Rabi Season
 - Perennial Season
 - Other Seasons
- **Total Gross Irrigated Area:** Combined irrigated area across all seasons
- **Net Irrigated Area:** Area irrigated at least once during the year
- **Groundwater Level (Pre-Monsoon):** Average groundwater level before monsoon
- **Groundwater Level (Post-Monsoon):** Average groundwater level after monsoon
- **Irrigation Scheme Type:** Indicates groundwater-based or surface water-based schemes

These attributes collectively enable comprehensive analysis of irrigation coverage and seasonal variations.

2.4 Data Types and Structure

The dataset contains a combination of numerical and categorical data types. Numerical attributes include area measurements and groundwater levels, while categorical attributes represent geographical names, scheme types, and administrative classifications.

The data is structured in a tabular format where:

- Each row represents a unique record corresponding to a village or administrative unit

- Each column represents a specific attribute

This structured format makes the dataset suitable for data preprocessing, modeling, and visualization using analytical tools.

2.5 Assumptions and Limitations of Data

While working with the dataset, certain assumptions have been made to ensure consistency and usability:

- All area-related values are assumed to be recorded in a uniform unit of measurement
- Missing or inconsistent values are treated during the preprocessing stage
- Seasonal data is assumed to follow standard agricultural seasons

The dataset also has some limitations:

- The data represents a fixed time period and does not include real-time updates
- Certain regional data may have missing or incomplete entries
- External factors such as climate variation are not directly included

Despite these limitations, the dataset is adequate for achieving the objectives of this project and for generating meaningful analytical insights.

3.DATA PREPROCESSING

3.1 Data Collection Method

Data preprocessing is a crucial step in any data analytics project, as the quality of input data directly impacts the accuracy and reliability of analysis and visualization. In this project, the dataset was obtained from a structured source related to irrigation and agricultural records. The data was collected in digital format and contained detailed information regarding geographical units, irrigation schemes, seasonal irrigation coverage, and groundwater levels.

Before analysis, the dataset was imported into the data analytics environment and examined to understand its structure, size, and content. Initial inspection helped identify potential issues such as missing values, inconsistencies, and redundant attributes that required preprocessing. The data collection process ensured that the dataset was comprehensive enough to support seasonal, district-wise, and village-wise analysis.

3.2 Data Cleaning

Data cleaning is the process of identifying and correcting errors, inconsistencies, and inaccuracies present in the raw dataset. Since real-world datasets are rarely perfect, cleaning the data was essential to ensure reliability and correctness of analysis.

3.2.1 Handling Missing Values

The dataset contained missing or null values in several attributes, particularly in numerical fields such as groundwater levels and irrigated areas. Missing values can distort analysis results and affect visualizations if not handled properly.

To address this issue:

- Records with insignificant missing data were retained after replacing missing values with appropriate statistical measures such as mean or median, depending on the nature of the attribute.
- In cases where entire records contained excessive missing information, those records were removed to maintain data integrity.
- For categorical attributes, missing values were replaced with suitable placeholders to preserve consistency.

This approach ensured that the dataset remained complete and usable without introducing bias or inaccuracies.

3.2.2 Handling Duplicate Records

Duplicate records were identified by comparing key attributes such as village name, block, and district. Duplicate entries can lead to incorrect aggregation and misleading insights.

The following steps were taken:

- Duplicate records were detected using unique identifiers and attribute comparisons.
- Redundant records were removed to prevent duplication in calculations and visualizations.
- The dataset was validated after duplicate removal to ensure no essential data was lost.

This step helped in maintaining the uniqueness of records and improving the accuracy of analysis.

3.3 Data Transformation

Data transformation involves converting data into a suitable format for analysis and visualization. This step ensures uniformity and enhances the usability of the dataset.

3.3.1 Data Formatting

Several formatting issues were addressed during this stage:

- Numerical values stored as text were converted into appropriate numerical data types.
- Inconsistent naming conventions in categorical fields were standardized.
- Units of measurement were verified and aligned across all relevant attributes.

Proper formatting enabled seamless calculations and aggregations during analysis.

3.3.2 Feature Selection

Not all attributes in the dataset were relevant to the project objectives. Feature selection was performed to retain only those attributes that contributed meaningful insights.

The selected features included:

- Geographical identifiers (state, district, block, village)
- Seasonal irrigated areas
- Net and gross irrigated areas
- Groundwater level indicators
- Irrigation scheme types

Irrelevant or redundant attributes were excluded to simplify the data model and improve dashboard performance.

3.4 Data Integration

In some cases, related data from multiple sources or tables needed to be integrated to create a unified dataset. Data integration ensured consistency and improved analytical depth.

Key steps involved:

- Combining tables using common keys such as district and village identifiers
- Verifying relationships between attributes
- Ensuring referential integrity across the dataset

This integration enabled comprehensive analysis across different dimensions such as geography and season.

3.5 Final Dataset Preparation

After completing data cleaning, transformation, and integration, the final dataset was prepared for modeling and visualization. This involved:

- Validating the dataset for accuracy and completeness
- Ensuring consistency across all attributes
- Testing the dataset by performing sample aggregations and visual checks

The finalized dataset was optimized for efficient data modeling and interactive dashboard creation. This clean and structured dataset served as a reliable foundation for subsequent stages of the project, including data modeling and visualization.

4.DATA MODELLING

4.1 Conceptual Data Model

The conceptual data model represents a high-level overview of the data structure and defines the main entities involved in the project along with their relationships. At this stage, the focus is on understanding *what data is required* and *how different data entities are related*, without considering implementation details.

In this project, the primary entities identified include geographical units (state, district, block, and village), irrigation schemes, irrigated areas, seasonal information, and groundwater levels. These entities form the backbone of the analytical framework and enable multidimensional analysis across different perspectives such as location and season.

The conceptual model helps in visualizing how irrigation data flows across different administrative levels and seasons, ensuring clarity before proceeding to logical and physical data modeling.

4.2 Logical Data Model

The logical data model translates the conceptual model into a structured format by defining tables, attributes, and relationships in more detail. It specifies how data elements are organized and connected while remaining independent of any specific software or tool.

In this project, the logical data model organizes data into multiple tables representing facts and dimensions. Each table includes clearly defined attributes, and relationships are established using primary and foreign keys. This structure ensures data consistency, reduces redundancy, and supports efficient querying and aggregation.

The logical data model serves as a blueprint for implementing the dataset within the data visualization tool.

4.3 Schema Design

To support effective analytical reporting and visualization, a **star schema** design approach was adopted. The star schema is widely used in analytical systems due to its simplicity, efficiency, and performance advantages.

4.3.1 Fact Table

The fact table stores quantitative measures related to irrigation and agricultural data. It acts as the central table in the schema and connects to multiple dimension tables.

Key measures in the fact table include:

- Gross irrigated area (season-wise)
- Net irrigated area
- Total irrigated area
- Cultivable area
- Net sown area
- Groundwater levels (pre-monsoon and post-monsoon)

Each record in the fact table corresponds to a specific geographical unit and season, enabling detailed analysis across multiple dimensions.

4.3.2 Dimension Tables

Dimension tables provide descriptive information that adds context to the numerical data stored in the fact table. The key dimension tables used in this project include:

- **Geographical Dimension:**
Contains state, district, block, and village details, enabling location-based analysis.
- **Season Dimension:**
Represents different agricultural seasons such as Kharif, Rabi, Perennial, and Other seasons.
- **Irrigation Scheme Dimension:**
Stores information related to irrigation scheme types, including groundwater-based and surface water-based schemes.

These dimension tables allow users to slice and filter data effectively in the dashboard.

4.4 Relationships and Keys

Relationships between fact and dimension tables were established using primary and foreign keys to maintain referential integrity.

- Each dimension table contains a unique primary key.
- The fact table includes foreign keys that reference corresponding dimension tables.
- Relationships were designed as one-to-many, ensuring efficient data aggregation and drill-down capabilities.

Proper relationship mapping ensures accurate calculations and seamless interactivity within the dashboard.

4.5 Model Validation

After designing the data model, validation was performed to ensure correctness and reliability. This involved:

- Verifying relationships between tables
- Testing aggregations and calculations
- Ensuring no circular dependencies or ambiguous relationships existed

The validated data model ensured optimal performance, accurate insights, and smooth user interaction in the visualization layer.

5. DASHBOARD DESIGN AND VISUALIZATIONS

5.1 Tools and Technologies Used

The dashboard for this project was developed using modern data visualization and business intelligence tools that support interactive and dynamic reporting. A data visualization platform was used to import the processed dataset, establish data relationships, and create interactive visuals.

The tool provides features such as data modeling, DAX-based calculations, slicers, filters, and interactive visual elements. These features enable users to explore data from multiple perspectives and gain meaningful insights with minimal effort. The use of visualization tools enhances data accessibility and simplifies complex datasets for better understanding.

5.2 Dashboard Layout and Theme

The dashboard layout was designed with a clear focus on usability, readability, and visual consistency. A structured layout was adopted to ensure that users can easily navigate through different insights without confusion.

Key design considerations included:

- A clean and professional color theme to maintain visual clarity
- Proper alignment and spacing of visuals
- Consistent font styles and sizes
- Logical grouping of related visuals

The dashboard theme was selected to reflect a professional and analytical appearance while ensuring that the data remains the primary focus.

5.3 Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) were included at the top of the dashboard to provide a quick summary of important metrics. These KPIs allow users to understand the overall status of irrigation coverage at a glance.

The KPIs used in the dashboard include:

- Total Gross Irrigated Area
- Net Irrigated Area
- Total Cultivable Area
- Net Sown Area
- Average Groundwater Level

These indicators help users quickly assess irrigation performance without needing to analyze detailed charts.

5.4 Visualization Types Used

Different visualization types were selected based on the nature of the data and the insights required.

5.4.1 Bar Charts

Bar charts were used to compare irrigated areas across districts and seasons. This visualization type is effective for showing variations and enabling easy comparison between categories.

5.4.2 Line Charts

Line charts were used to represent trends in irrigation data across seasons. They help visualize changes over time and identify seasonal patterns.

5.4.3 Pie and Donut Charts

Pie and donut charts were used to represent the proportion of irrigation scheme types, such as groundwater-based and surface water-based schemes. These visuals provide a clear understanding of percentage distribution.

5.4.4 Cards and Tables

Cards were used to display key metrics and summary values, while tables provided detailed views of village-wise or district-wise data. These visuals enhance both overview and detailed analysis.

5.5 Filters and Slicers

Filters and slicers were implemented to enhance interactivity and allow users to explore data dynamically. Users can filter data based on:

- District
- Village
- Season
- Irrigation scheme type

Slicers enable users to drill down into specific subsets of data, making the dashboard more flexible and user-friendly.

5.6 User Interactivity

User interactivity was a core design consideration. The dashboard supports cross-filtering, enabling users to click on one visual and automatically update others. Tooltips were also included to display additional information when hovering over data points.

These interactive features improve user engagement and help users derive insights efficiently.

6.ANALYSIS AND INSIGHTS

6.1 Overall Data Analysis

The analysis performed through the interactive dashboard provides a comprehensive overview of irrigation patterns across different geographical regions and agricultural seasons. By integrating cleaned and well-structured data into the dashboard, it becomes possible to examine variations in irrigated area, scheme distribution, and groundwater levels in an intuitive manner.

The overall analysis indicates noticeable differences in irrigation coverage among districts and villages. Some regions demonstrate consistently higher irrigated areas due to better availability of water resources and irrigation infrastructure, while others show limited coverage, highlighting potential areas for improvement.

The dashboard effectively consolidates large volumes of data into meaningful visuals, enabling users to identify patterns that would otherwise be difficult to detect using raw tabular data.

6.2 Trend Analysis

Trend analysis was conducted to study variations in irrigation patterns across different agricultural seasons. Seasonal trends reveal that irrigation demand and coverage differ significantly between Kharif, Rabi, and other seasons.

The analysis shows that:

- Kharif season generally records higher irrigated area due to dependence on monsoon rainfall.
- Rabi season irrigation is comparatively lower in certain regions and relies more on groundwater and artificial irrigation methods.
- Perennial irrigation remains consistent throughout the year but is limited to specific areas with reliable water sources.

These trends provide valuable insights into seasonal water dependency and highlight the importance of effective irrigation planning to ensure year-round agricultural productivity.

6.3 Comparative Analysis

Comparative analysis was carried out to evaluate differences between regions, seasons, and irrigation scheme types. District-wise comparison reveals that some districts have significantly higher net irrigated areas compared to others, indicating unequal distribution of irrigation facilities.

The comparison between groundwater-based and surface water-based irrigation schemes shows varying dependency patterns across regions. Areas with limited surface water sources rely heavily on groundwater, which may pose sustainability concerns if not managed properly.

Such comparative insights help stakeholders understand regional disparities and prioritize resource allocation accordingly.

6.4 Key Observations

Based on the analysis performed using the dashboard, several important observations were identified:

- There is a noticeable variation in irrigated area across districts and villages.
- Seasonal irrigation patterns are strongly influenced by monsoon availability.
- Groundwater dependency is higher in regions with limited surface water sources.
- Some regions demonstrate underutilization of cultivable land due to insufficient irrigation coverage.
- Post-monsoon groundwater levels are generally higher than pre-monsoon levels, indicating seasonal recharge.

These observations highlight both strengths and challenges within the existing irrigation framework.

6.5 Insights Derived from Dashboard

The dashboard enables stakeholders to derive actionable insights through interactive exploration. Users can quickly identify high-performing and low-performing regions, analyze seasonal dependencies, and assess groundwater sustainability.

The insights derived from this analysis can support:

- Better irrigation planning and management
- Identification of regions requiring infrastructure development
- Seasonal resource optimization
- Data-driven policy formulation

Overall, the analysis demonstrates the effectiveness of data visualization in transforming complex irrigation data into clear, actionable insights.

7.CONCLUSION

7.1 Summary of Findings

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This Minor Project focused on analyzing irrigation-related data using structured data preprocessing, efficient data modeling, and interactive dashboard visualization techniques. The project successfully transformed raw and complex datasets into meaningful visual insights that highlight irrigation patterns across different regions and seasons.

Through the dashboard, variations in irrigated areas, seasonal dependencies, and groundwater usage were clearly identified. The analysis revealed significant differences in irrigation coverage among districts and villages, indicating unequal access to irrigation resources. Seasonal trends demonstrated the strong influence of monsoon rainfall on irrigation patterns, particularly during the Kharif season

7.2 Achievement of Objectives

All the objectives defined at the beginning of the project were successfully achieved. The dataset was thoroughly cleaned and preprocessed to ensure accuracy and consistency. A well-structured data model was designed to support efficient analysis, and an interactive dashboard was developed to visualize key metrics and trends.

The project effectively demonstrated how data analytics and visualization tools can be used to analyze large datasets and extract meaningful insights. The dashboard allows users to explore data dynamically and supports informed decision-making.

7.3 Learning Outcomes

This project provided valuable hands-on experience in data preprocessing, data modeling, and dashboard design. It enhanced understanding of analytical workflows and strengthened problem-solving and critical thinking skills. The project also improved familiarity with data visualization tools and reinforced the importance of data-driven decision-making in real-world applications.

Final Dashboard



8. FUTURE SCOPE

Although the project achieved its intended objectives, there are several opportunities for further enhancement and extension in the future.

8.1 Scope for Data Expansion

The dataset can be expanded by including data from multiple years to enable time-series analysis. This would help in understanding long-term irrigation trends and changes over time.

8.2 Advanced Analytics Integration

Future work may incorporate predictive analytics and machine learning techniques to forecast irrigation demand, groundwater depletion, and seasonal water requirements.

8.3 Automation and Real-Time Dashboards

The dashboard can be enhanced by integrating real-time data sources, enabling automated updates and real-time monitoring of irrigation patterns.

8.4 Enhancements in Visualization

Additional advanced visualizations such as geospatial maps and drill-through reports can be incorporated to improve analytical depth and user experience.

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